Main Experiment Tables with Local Reproduction Rows

Reproduction summary

1 PruneVid

Method	Retained	FLOPs	MVBench	VideoMME	EgoSchema (Subset / Fullset)
PLLaVA	100.0%	1.00×	46.6	44.4	47.8 / 42.6
PLLaVA w/ FastV	30.0%	0.33×	46.1	43.6	46.2 / 41.0
PLLaVA w/ Prumerge	55.7%	0.53×	45.6	43.8	45.2 / 40.4
PLLaVA w/ Look-M	20.0%	1.00×	46.6	44.3	47.0 / 42.3
PLLaVA w/ Ours	**16.2%**	**0.23**×	**47.6**	**45.0**	**49.0** / **42.6**
PLLaVA w/ Ours (Reproduce)	16.2%	0.23×	46.42	44.89	48.40 / 41.90
ST-LLM	100.0%	1.00×	54.9	42.0	56.2 / 45.6
ST-LLM w/ FastV	30.0%	0.37×	42.9	34.5	48.0 / 38.5
ST-LLM w/ Look-M	20.0%	1.00×	54.0	40.6	54.0 / 44.5
ST-LLM w/ Ours	**15.1%**	**0.26**×	**54.3**	**41.4**	**54.6** / **44.7**
ST-LLM w/ Ours (Reproduce, self-port)	15.1%	0.26×	54.80	42.59	59.80 / 45.10
LLaVA-OneVision	100.0%	1.00×	58.0	58.2	62.0 / 60.0
LLaVA-OneVision w/ FastV	30.0%	0.30×	57.2	57.6	62.6 / 60.0
LLaVA-OneVision w/ Prumerge	55.2%	0.49×	52.9	56.7	62.2 / 60.0
LLaVA-OneVision w/ Look-M	20.0%	1.00×	57.0	58.0	62.0 / **59.8**
LLaVA-OneVision w/ Ours	**17.0%**	**0.20**×	**57.5**	**58.6**	**62.6** / 59.5
LLaVA-OneVision w/ Ours (Reproduce, self-port)	17.0%	0.20×	57.23	58.37	64.40 / 61.78

Table 1: PruneVid main experiment table with local reproduction rows inserted below the paper Ours rows. PLLaVA uses the released implementation; ST-LLM and LLaVA-OneVision are self-ported because the official repository marks those PruneVid backbones as not yet released.

2 VidCom^2

Backbone	Method / Ratio	MVBench	LongVideoBench	MLVU	VideoMME	Short	Medium	Long	Average (%)
LLaVA-OV-7B	Upper bound	56.9	56.4	63.0	58.6	70.3	56.6	48.8	100.0
LLaVA-OV-7B	DyCoke, $R = 30\%$	56.6	54.7	60.3	56.1	67.1	54.6	46.6	96.5
LLaVA-OV-7B	Random, $R = 25\%$	54.2	52.7	59.7	55.6	65.4	53.0	48.3	94.8
LLaVA-OV-7B	FastV, $R = 25\%$	55.5	53.3	59.6	55.3	65.0	53.8	47.0	94.9
LLaVA-OV-7B	PDrop, $R = 25\%$	55.3	51.3	57.1	55.5	64.7	53.1	48.7	94.1
LLaVA-OV-7B	SparseVLM, $R = 25\%$	56.4	53.9	60.7	57.3	68.4	55.2	48.1	97.5
LLaVA-OV-7B	DyCoke, $R = 25\%$	49.5	48.1	55.8	51.0	61.1	48.6	43.2	87.0
LLaVA-OV-7B	**VidCom^2**, $R = 25\%$	**57.2**	**54.9**	**62.5**	**58.6**	**69.8**	**56.4**	**49.4**	**99.6**
LLaVA-OV-7B	VidCom^2, $R = 25\%$ (Reproduce)	57.00	55.27	62.09	58.33	69.33	56.56	49.11	99.3
LLaVA-OV-7B	FastV, $R = 15\%$	51.6	48.3	55.0	48.1	51.4	49.4	43.3	85.0
LLaVA-OV-7B	PDrop, $R = 15\%$	53.2	47.6	54.7	50.1	58.7	48.7	45.0	87.4
LLaVA-OV-7B	SparseVLM, $R = 15\%$	52.9	49.7	57.4	53.4	61.0	52.1	47.0	91.2
LLaVA-OV-7B	**VidCom^2**, $R = 15\%$	**54.3**	**52.0**	**58.9**	**56.2**	**65.8**	**54.8**	**48.1**	**95.1**
LLaVA-OV-7B	VidCom^2, $R = 15\%$ (Reproduce)	54.48	53.10	60.47	55.74	65.56	53.78	47.89	95.2
LLaVA-Video-7B	Upper bound	60.4	59.6	70.3	64.3	77.2	62.1	53.4	100.0
LLaVA-Video-7B	DyCoke, $R = 30\%$	57.5	55.5	60.6	61.3	73.4	59.3	51.2	93.8
LLaVA-Video-7B	FastV, $R = 25\%$	53.8	51.2	57.8	59.3	67.1	60.0	**50.8**	89.7
LLaVA-Video-7B	SparseVLM, $R = 25\%$	55.4	54.2	58.9	60.1	71.1	59.1	50.1	91.6
LLaVA-Video-7B	DyCoke, $R = 25\%$	50.8	53.0	56.9	56.1	65.8	53.6	48.9	86.3
LLaVA-Video-7B	**VidCom^2**, $R = 25\%$	**57.0**	**55.5**	**59.0**	**61.7**	**73.0**	**61.7**	50.0	**93.6**
LLaVA-Video-7B	VidCom^2, $R = 25\%$ (Reproduce)	56.88	56.62	57.24	61.85	72.89	61.67	51.00	93.5
LLaVA-Video-7B	FastV, $R = 15\%$	44.0	44.6	53.8	51.3	56.4	51.1	46.2	78.0
LLaVA-Video-7B	SparseVLM, $R = 15\%$	53.1	**52.7**	56.2	55.7	65.0	53.9	48.3	86.3
LLaVA-Video-7B	**VidCom^2**, $R = 15\%$	**53.3**	51.5	**56.8**	**58.3**	**68.0**	**57.3**	**49.7**	**88.5**
LLaVA-Video-7B	VidCom^2, $R = 15\%$ (Reproduce)	53.33	52.28	54.94	58.22	68.00	56.89	49.78	88.0

Table 2: VidCom^2 main experiment table with local LLaVA-OV and LLaVA-Video reproduction rows inserted below the corresponding paper rows. Average(%) follows the paper's convention: the row's 7-column sum (MVBench, LongVideoBench, MLVU, VideoMME, Short/Medium/Long) divided by the corresponding Upper-bound row's 7-column sum. Reproduced Average(%) matches the paper within ≤ 0.5 pts on all four rows (OV R=25%: 99.3 vs 99.6; OV R=15%: 95.2 vs 95.1; Video R=25%: 93.5 vs 93.6; Video R=15%: 88.0 vs 88.5).

3 AOT

Method	Prefill FLOPs	FLOPs Ratio	Retained	MVBench	EgoSchema	LongVideoBench	VideoMME	Avg. Score	Avg. %
LLaVA-OV-7B	40.8	100%	100%	58.3	60.4	56.4	58.6	58.4	100.0
FastV	9.3	22.8%	100%	55.9	57.5	**56.7**	56.1	56.5	96.7
PDrop	10.5	25.7%	100%	56.1	58.0	54.1	56.4	56.2	96.2
DyCoke	8.7	21.3%	25%	53.1	59.5	49.5	54.3	54.1	92.6
VisionZip	8.7	21.3%	25%	57.9	60.3	56.5	58.2	58.2	99.7
PruneVid	8.7	21.3%	25%	57.4	59.9	55.7	57.4	57.6	98.6
FastVID	8.7	21.3%	25%	56.5	–	56.3	58.0	–	–
AOT	8.7	21.3%	25%	**58.7**	**61.3**	56.3	57.5	**58.5**	**100.0**
AOT (Reproduce)	8.7	21.3%	25%	57.75	61.18	54.75	54.89	57.14	97.8
VisionZip	7.0	17.2%	20%	57.7	59.8	55.2	57.9	57.7	98.8
PruneVid	7.0	17.2%	20%	57.2	59.7	54.7	56.9	57.1	97.8
FastVID	7.0	17.2%	20%	56.3	–	**57.1**	**57.9**	–	–
AOT	7.0	17.2%	20%	**58.1**	**61.3**	56.2	57.2	**58.2**	**99.7**
AOT (Reproduce)	7.0	17.2%	20%	57.73	61.22	54.08	54.78	56.95	97.5
VisionZip	5.2	12.7%	15%	56.5	59.8	54.4	56.1	56.7	97.1
PruneVid	5.2	12.7%	15%	56.8	59.7	55.4	56.6	57.1	97.8
FastVID	5.2	12.7%	15%	56.0	–	**56.2**	**57.7**	–	–
AOT	3.4	8.3%	15%	**57.8**	**61.3**	55.2	56.6	**57.7**	**98.8**
AOT (Reproduce)	3.4	8.3%	15%	57.33	60.98	52.80	54.22	56.33	96.5
VisionZip	3.4	8.3%	10%	53.5	58.0	49.3	53.4	53.5	91.6
PruneVid	3.4	8.3%	10%	56.2	59.8	54.5	56.0	56.6	96.9
FastVID	3.4	8.3%	10%	55.9	–	**56.3**	**57.3**	–	–
AOT	5.2	12.7%	10%	**57.0**	**60.6**	54.2	56.1	**57.0**	**97.6**
AOT (Reproduce)	5.2	12.7%	10%	56.70	60.45	52.65	53.78	55.90	95.7

Table 3: AOT main experiment table on LLaVA-OneVision with local reproduction rows inserted below the corresponding paper AOT rows. All runs follow the authors' released evaluation config (`eval_ov-7b.sh`); EgoSchema full-test accuracy is obtained by submitting local predictions to the official validation server; LongVideoBench and VideoMME use a VTN-calibrated sweep so measured token retention matches the labeled ratios. Averages are computed over the four benchmarks, with Avg. % relative to the full-model baseline.

Method	Prefill FLOPs	FLOPs Ratio	Retained	MVBench	EgoSchema	LongVideoBench	VideoMME	Avg. Score	Avg. %
LLaVA-Video-7B	80.2	100%	100%	60.4	57.2	58.9	64.3	60.2	100.0
FastV	17.1	21.3%	100%	54.3	54.1	55.0	58.8	55.6	92.4
PDrop	19.5	24.3%	100%	55.9	54.3	54.7	61.9	56.7	94.2
VisionZip	9.3	18.9%	25%	56.7	54.7	54.7	60.7	56.7	94.2
DyCoke	9.3	18.9%	25%	50.8	–	53.0	56.9	–	–
AOT	9.3	18.9%	25%	**58.8**	**55.4**	**56.2**	**62.4**	**58.2**	**96.7**
AOT (Reproduce)	9.3	18.9%	25%	58.42	55.85‡	56.24	62.52	58.26	96.8
VisionZip	9.3	11.6%	15%	56.7	54.7	54.7	60.7	56.7	94.2
AOT	9.3	11.6%	15%	**57.8**	**55.2**	**55.0**	**62.0**	**57.5**	**95.5**
AOT (Reproduce)	9.3	11.6%	15%	57.98	55.08‡	55.42	62.11	57.65	95.8

Table 4: AOT main experiment table on LLaVA-Video with local reproduction rows inserted below the corresponding paper AOT rows. ‡EgoSchema full-test labels are not public; accuracy is obtained by submitting the local predictions to the official validation server. Averages are computed over the four benchmarks, with Avg. % relative to the full-model baseline.